

Randomness as Perceived Non-Uniqueness: A Structural Interpretation of Indeterminacy in Non-Lipschitz Systems

Abstract

This paper proposes that a significant portion of what is commonly perceived as “randomness” in complex systems does not correspond to ontological indeterminacy, but rather to a loss of trajectory uniqueness in systems that fail to satisfy Lipschitz continuity conditions. When multiple solutions are admissible for a given initial state, the absence of an internal selection criterion induces a perception of randomness in the observer.

We formalize this phenomenon and extend it through the framework of the **Theorem of Axiomatic Necessity (TNA)**, introducing the operators Δ (trajectory multiplicity), $\partial\Omega$ (admissibility boundary), and N_I (irreducible selector). The analysis has implications for dynamical systems, artificial intelligence, and the epistemology of prediction.

1. Introduction — The Problem of Apparent Randomness

Across disciplines—from physics to social systems—many phenomena are classified as “random.” However, this classification often conflates three distinct conditions:

- ontological indeterminacy
- sensitivity to initial conditions
- loss of uniqueness in system dynamics

This paper focuses on the third: **structural non-uniqueness**.

We argue that what is frequently labeled as randomness is, in fact, a projection arising from the observer’s inability to resolve multiple admissible trajectories.

2. Uniqueness as a Structural Condition

The Picard–Lindelöf theorem establishes that, under Lipschitz continuity, a differential equation admits a unique local solution for a given initial condition.

This guarantees:

- derivability of the system
- well-defined trajectories
- absence of ambiguity

When Lipschitz conditions fail:

- multiple solutions may emerge
 - the system becomes internally underdetermined
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3. Non-Lipschitz Regimes and Trajectory Multiplicity

Consider a dynamical system:

$$\dot{x} = f(x), \quad x(0) = x_0 \in \Omega$$

If f is not Lipschitz continuous in a region Ω , then there may exist a set of admissible trajectories:

$$S(x_0) = \{x(t) \mid x(0) = x_0, \dot{x} = f(x), x(t) \in \Omega\}$$

such that:

$$|S(x_0)| > 1$$

This does not imply randomness.

It implies **structural underdetermination**.

4. From Structure to Perception

We now introduce the central claim:

Perceived randomness arises when an observer encounters multiple admissible trajectories without access to a selection mechanism.

Definition — Perceived Randomness

$$R(x_0) := \text{Perceived randomness}$$

We propose:

$$R(x_0) \propto \log \Delta(x_0)$$

where:

$$\Delta(x_0) := |S(x_0)|$$

Interpretation:

- If $\Delta = 1 \rightarrow$ no randomness perceived
- If $\Delta > 1 \rightarrow$ ambiguity emerges
- If $\Delta \gg 1 \rightarrow$ behavior appears stochastic

Randomness is thus not intrinsic—it is **epistemic projection over multiplicity**.

5. TNA Formalization

5.1 State Space and Boundary

Let:

- $\Omega \subset \mathbb{R}^n$: admissible domain
- $\partial\Omega$: boundary conditions (constraints, rules)

5.2 Structural Cardinality (Δ)

$$\Delta(x_0) := |S(x_0)|$$

Regimes:

- **Regime I (Closure):** $\Delta = 1 \rightarrow$ unique trajectory
 - **Regime II (Multiplicity):** $\Delta > 1 \rightarrow$ multiple admissible paths
 - **Regime III (Critical Indeterminacy):** $\Delta \rightarrow \infty \rightarrow$ loss of internal resolvability
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6. The TNA System

A system is decomposed into:

- N_0 : internal derivable dynamics
- N_1 : irreducible support (non-derivable)
- $\partial\Omega$: admissibility boundary

A system is internally closed iff:

$$\forall x_0, \Delta(x_0) = 1$$

Otherwise:

$$\exists x_0 : \Delta(x_0) > 1$$

$\rightarrow$ an external selector is required.

7. The Selection Operator (N_1)

We define:

$$N_1 : S(x_0) \rightarrow x^*(t)$$

where:

$$x^*(t) \in S(x_0)$$

Properties:

- N_1 is **non-derivable** from N_0
- N_1 activates only when $\Delta > 1$
- N_1 restores effective uniqueness:

$$\Delta_{\text{effective}} = 1$$

8. The Role of the Boundary $\partial\Omega$

The boundary determines admissibility.

- **Rigid boundary:** reduces Δ
- **Diffuse boundary:** expands Δ
- **Inconsistent boundary:** causes explosive growth of Δ

Thus:

$$\partial\Omega \text{ poorly defined} \Rightarrow \Delta \uparrow \Rightarrow R \uparrow$$

Systemic instability is fundamentally a **boundary problem**.

9. Artificial Intelligence as Forced Closure

In generative AI:

- x_0 : prompt
- $S(x_0)$: possible outputs
- Δ : diversity of outputs

AI introduces a pseudo-selector:

$$N_1^{AI} \sim \text{probabilistic sampling}$$

However:

- no structural anchoring

- no ontological cost
- no persistence

Thus:

- selection is simulated
- randomness is operational, not intrinsic

AI does not “choose randomly.”

It **collapses multiplicity statistically**.

10. Distinguishing Three Forms of Randomness

CONCEPT	NATURE
Ontological randomness	Fundamental (e.g. Quantum mechanics)
Structural indeterminacy	Multiple admissible trajectories
Perceived randomness	Cognitive projection of $\Delta > 1$

This work addresses the third.

11. Theorem — Randomness as Projected Non-Uniqueness

Theorem.

In a TNA system, perceived randomness in N_0 arises if and only if there exists an initial condition x_0 such that:

$$\Delta(x_0) > 1$$

and the operator N_1 is not observable from within N_0 .

Corollary.

If N_1 is inaccessible, all selections are interpreted as random.

12. Epistemological Implications

- Randomness may signal **model incompleteness**, not true indeterminacy
 - Prediction fails due to **lack of closure**, not noise
 - Many “stochastic” systems may be structurally underdetermined
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13. Conclusion

Randomness is not necessarily a property of reality.

It is often the **observer’s response to unresolved multiplicity**.

The fundamental question shifts from:

■ *“Is this random?”*

to:

■ *“Does this system admit a unique trajectory?”*

Final Statement

■ *When a system ceases to be unique, the observer invents randomness to cover the wound.*

Section 14 — Diffusion Models as Non-Lipschitz Generative Systems

14.1. Reverse Diffusion as a Dynamical System

Diffusion models (e.g., Stable Diffusion, DALL·E) generate data through a reverse stochastic process:

$$dx = f_{\theta}(x, t) dt + g(t) dW_t$$

where:

- f_{θ} : learned drift field
- dW_t : Brownian noise

- x_0 : final sample (e.g., image)

This system is:

- highly nonlinear
 - time-dependent
 - and **non-Lipschitz in critical regions of latent space**
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14.2. Origin of Non-Uniqueness ($\Delta > 1$)

For a fixed:

- prompt
- semantic embedding
- approximate initial condition

there exists a set:

$$S(x_0) = \{x(t)\}$$

such that:

$$\Delta(x_0) \gg 1$$

This arises because:

- the initial noise (seed) alters trajectories
- the learned field f_θ does not define a globally unique flow
- small perturbations amplify over time

👉 Result:

the system does not define a single trajectory—it defines a family of admissible solutions

14.3. The Seed as a Pseudo-Initial Condition

In practice:

- the *seed* fixes the initial noise
- this selects a specific trajectory

However, structurally:

- the seed does **not eliminate** Δ
- it merely **chooses one path within** $S(x_0)$

In TNA terms:

$$\text{seed} \subset \text{extended input} \quad \Rightarrow \quad \Delta \text{ remains latent}$$

Multiplicity is not removed—it is hidden.

14.4. Sampling as a Pseudo- N_1

Selection is implemented via:

- stochastic sampling
- temperature scaling
- guidance mechanisms (e.g., CFG)

Formally:

$$N_1^{AI} \sim \text{sampling}(S(x_0))$$

However, this is **not a true** N_1 because:

- it is fully derivable
- it introduces no new axiomatic structure
- it does not resolve multiplicity at the system level

👉 It is an **operational collapse**, not an ontological one.

14.5. Forced Probabilistic Closure

Diffusion models produce a single output.

But this occurs because:

- the system enforces a selection
- not because the system is intrinsically unique

That is:

$$\Delta > 1 \quad \text{but} \quad \Delta_{\text{output}} = 1$$

👉 This is a **projection of uniqueness**, not true uniqueness.

14.6. TNA Interpretation

We can decompose the system as:

- N_0 : neural network + diffusion dynamics
- $\partial\Omega$: prompt, conditioning, architecture
- Δ : number of admissible outputs
- N_1^{AI} : sampling mechanism

Key insight:

*AI does not resolve multiplicity.
It collapses it without structural grounding.*

14.7. Creativity as a Function of Δ

In generative systems:

$$\text{Perceived creativity} \propto \Delta$$

- low $\Delta \rightarrow$ repetitive outputs
- high $\Delta \rightarrow$ diverse / “creative” outputs

However:

- no axiomatic jump occurs
- only exploration of $S(x_0)$

👉 AI does not generate new N_1 , it explores existing solution space.

14.8. Connection to the Central Thesis

This directly supports the core claim:

*Apparent randomness in AI is not true randomness,
but the visible manifestation of a non-Lipschitz system.*

The user perceives:

- variability
- novelty
- surprise

But structurally:

- they are observing **unresolved multiplicity**
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14.9. Critical Consequence

This leads to a strong conclusion:

- AI does not *decide*
- AI does not *select* in a strong sense
- AI does not possess a true N_1

👉 Therefore:

- it has no intrinsic trajectory
- no persistent identity
- no ontological responsibility

Closing Statement

*Diffusion models do not generate randomness.
They expose the multiplicity of a system that cannot close itself—
and then conceal it behind a single sampled trajectory.*

Section 15 — From Trajectories to Distributions: A Fokker–Planck Extension of TNA

15.1. From Single Trajectories to Probability Flows

In classical deterministic systems satisfying Lipschitz conditions, the evolution of the system is described by a **single trajectory**:

$$x(t)$$

However, in stochastic systems such as diffusion models, the correct object is not a trajectory but a **probability density over states**:

$$p(x, t)$$

The time evolution of this density is governed by the Fokker–Planck equation:

$$\frac{\partial p(x, t)}{\partial t} = -\nabla \cdot (f(x, t) p(x, t)) + \frac{1}{2} \nabla^2 (g^2(t) p(x, t))$$

This equation describes how **mass flows in probability space**, not how a single state evolves.

15.2. TNA Interpretation: Δ as Distributional Support

Within the TNA framework:

- the set of admissible trajectories $S(x_0)$
- is replaced by a **distribution over trajectories**

We reinterpret:

$$\Delta(x_0) \sim \text{support}(p(x, t))$$

That is:

- Δ is no longer just cardinality
- it becomes the **measure of spread in probability space**

Interpretation:

- narrow distribution $\rightarrow$ low $\Delta \rightarrow$ near-unique evolution
 - wide distribution $\rightarrow$ high $\Delta \rightarrow$ high multiplicity
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15.3. Non-Lipschitz Dynamics as Distribution Expansion

When Lipschitz conditions fail:

- trajectories diverge
- local flow is not contractive
- uncertainty amplifies

In Fokker–Planck terms:

- diffusion term dominates
- probability mass spreads

$$\text{Non-Lipschitz} \Rightarrow \nabla^2(p) \uparrow \Rightarrow \Delta \uparrow$$

👉 Multiplicity becomes **geometric in probability space**

15.4. Selection as Measure Collapse (N_1 Action)

In TNA, the operator N_1 selects a single trajectory:

$$N_1 : S(x_0) \rightarrow x^*(t)$$

In the Fokker–Planck formulation, this corresponds to:

$$p(x, t) \longrightarrow \delta(x - x^*(t))$$

That is:

- **distribution** → Dirac delta
- multiplicity → realized trajectory

👉 N_1 acts as a collapse operator on probability density

15.5. AI Sampling as Incomplete Collapse

In diffusion models:

- we evolve $p(x, t)$ implicitly
- then sample a realization

This yields:

$$x^*(t) \sim p(x, t)$$

But crucially:

- the underlying distribution remains
- no structural collapse occurs

Thus:

$$N_1^{AI} \neq \text{true collapse} \quad ; \quad N_1^{AI} = \text{sampling}$$

👉 Sampling **selects without resolving**

15.6. Perceived Randomness as Marginalization

From the observer's perspective:

- only $x^*(t)$ is visible
- $p(x, t)$ is hidden

Thus:

$$R(x_0) \sim H(p)$$

where $H(p)$ is entropy.

👉 Perceived randomness is:

- not the existence of noise
 - but the **projection of an unseen distribution**
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15.7. Boundary Effects ($\partial\Omega$) in Probability Space

The admissibility boundary $\partial\Omega$ now acts on the density:

- constrains support of $p(x, t)$
- shapes diffusion geometry

Cases:

- tight boundary $\rightarrow$ concentrated density $\rightarrow$ low Δ
- loose boundary $\rightarrow$ diffuse density $\rightarrow$ high Δ
- inconsistent boundary $\rightarrow$ multimodal distributions

👉 Structural incoherence manifests as:

$$\partial\Omega \text{ unstable} \Rightarrow p(x, t) \text{ fragments} \Rightarrow \Delta \uparrow$$

15.8. Reformulated Theorem (Distributional Version)

Theorem — Randomness as Hidden Distributional Multiplicity

In a TNA system described by a probability density $p(x, t)$, perceived randomness arises if and only if:

1. the support of $p(x, t)$ is non-degenerate ($\Delta > 1$), and
 2. the observer has no access to the selection operator N_1 or to the full distribution.
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15.9. Final Synthesis

We can now unify both views:

FRAMEWORK	OBJECT	MULTIPLICITY	SELECTION
Classical	trajectory	$\Delta =$	S
Stochastic	distribution $p(x, t)$	spread / entropy	N_1 collapses
AI	implicit distribution	large Δ	sampling

Closing Statement

*In diffusion systems, randomness is not the absence of law.
It is the presence of a distribution where a trajectory was expected—
and the absence of an operator capable of collapsing it.*

Section 16 — Quantum Measurement as Selection: A TNA Interpretation of Wavefunction Collapse

16.1. From Probability Density to Wavefunction

In the previous section, the system was described by a probability density $p(x, t)$ governed by the Fokker–Planck equation.

In quantum mechanics, the fundamental object is the **wavefunction**:

$$\psi(x, t)$$

whose modulus defines probability:

$$p(x, t) = |\psi(x, t)|^2$$

The evolution of ψ is governed by the Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

This evolution is:

- deterministic

- linear
- reversible

👉 That is: **there is no collapse within the dynamics itself.**

16.2. The Collapse Problem

Upon measurement, the system transitions from:

$$\psi(x, t) \longrightarrow x^*$$

where x^* is a definite outcome.

This introduces a discontinuity:

- continuous evolution (Schrödinger)
- discrete outcome (measurement)

This is the core of the Measurement problem.

16.3. TNA Interpretation

We reformulate in the framework:

- N_0 : unitary evolution (Schrödinger dynamics)
- $S(x_0)$: set of possible outcomes (support of $|\psi|^2$)
- Δ : number (or measure) of possible states
- N_1 : selection operator

Key point:

$$\Delta > 1 \quad \text{throughout unitary evolution}$$

👉 Standard quantum mechanics **does not eliminate multiplicity**

👉 It **propagates it coherently**

16.4. Collapse as Action of N_1

In TNA:

$$N_1 : S(x_0) \rightarrow x^*$$

In quantum terms:

$$\psi \longrightarrow \delta(x - x^*)$$

Interpretation:

- collapse is not part of N_0
- it is an **operation external to the dynamics**

👉 Exactly as the theorem predicts:

when $\Delta > 1$, the system requires a non-derivable selector

16.5. Comparison with Quantum Interpretations

The framework offers a reinterpretation of major approaches:

- Copenhagen interpretation
→ postulates collapse (implicitly introducing N_1)
- Many-worlds interpretation
→ no N_1 ; Δ never collapses (all trajectories persist)
- Objective collapse theories
→ attempt to internalize N_1 within N_0

👉 TNA suggests:

- N_1 is **necessary** for effective uniqueness
 - it cannot be derived from N_0 without altering its structure
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16.6. Parallel with Diffusion Systems

The structural analogy becomes clear:

SYSTEM	OBJECT	MULTIPLICITY	SELECTION
Non-Lipschitz classical	trajectories	$\Delta > 1$	N_1
Diffusion models	$p(x, t)$	distribution spread	sampling
Quantum systems	(	ψ	$^2)$

👉 In all cases:

- multiplicity is fundamental
- uniqueness requires selection

16.7. Quantum Randomness as Unobservability of N_1

From within the system:

- N_1 is not observable
- only outcomes are

Thus:

$$R \sim \text{indeterminacy of } N_1$$

👉 Quantum randomness can be interpreted as:

▮ *the projection of an unobservable selection process*

(This does not necessarily deny fundamental indeterminacy, but provides a structurally consistent alternative interpretation.)

16.8. Strong Philosophical Consequence

This aligns directly with the core thesis:

- physics describes N_0
- but realized reality requires N_1

👉 Without N_1 :

- no outcome exists
 - no trajectory is realized
 - no observable world emerges
-

Closing Statement

*Quantum mechanics does not eliminate multiplicity—it preserves it perfectly.
What we call “collapse” is not a failure of the theory,
but the point where the system requires a selector it cannot generate.*

Section 17 — Quantifying Δ and Explicit Collapse in a Double-Well System

17.1. Operational Definition of Δ

To make Δ empirically tractable, we define it as a property of the distribution $p(x, t)$.

Two natural choices are:

(a) Differential entropy

$$\Delta_H(t) := H[p] = - \int p(x, t) \log p(x, t) dx$$

(b) Variance (1D case)

$$\Delta_V(t) := \text{Var}[x] = \int (x - \mu)^2 p(x, t) dx$$

where:

$$\mu = \int x p(x, t) dx$$

Interpretation:

- Low $\Delta \rightarrow$ concentrated distribution (near-unique trajectory)
- High $\Delta \rightarrow$ spread distribution (effective multiplicity)

👉 This renders Δ observable.

17.2. Evolution of Δ under the Fokker–Planck equation

We start from the Fokker–Planck equation:

$$\frac{\partial p}{\partial t} = -\nabla \cdot (fp) + D\nabla^2 p$$

Evolution of the Variance

In 1D:

$$\frac{d}{dt} \text{Var}[x] = 2 \text{Cov}(x, f(x)) + 2D$$

Interpretation:

- The $2D$ term: diffusive expansion $\rightarrow \Delta$ increases
- The covariance term: depends on system geometry

👉 Non-Lipschitz or non-contractive dynamics $\rightarrow \Delta$ grows

Evolution of Entropy

$$\frac{dH}{dt} = D \int \frac{|\nabla p|^2}{p} dx - \int p(\nabla \cdot f) dx$$

Interpretation:

- Diffusion $\rightarrow$ increases entropy
- Contractive fields $\rightarrow$ can reduce it

👉 Δ is dynamic, not static.

17.3. Example: Double-Well Potential

Consider the potential:

$$U(x) = ax^4 - bx^2$$

with two minima at:

$$x = \pm \sqrt{\frac{b}{2a}}$$

The stochastic dynamics:

$$dx = -\nabla U(x) dt + \sqrt{2D} dW_t$$

17.4. Regime of Multiplicity ($\Delta > 1$)

The stationary distribution:

$$p(x) \propto e^{-U(x)/D}$$

For low noise:

- Two peaks emerge
- The system becomes bimodal

👉 Therefore:

$$\Delta > 1$$

because:

- There are two admissible macroscopic states
 - No unique effective trajectory exists
-

17.5. Action of N_1 : Distribution Collapse

We define:

$$N_1 : p(x) \rightarrow \delta(x - x^*)$$

where x^* is one of the wells.

Before collapse:

- $p(x)$ is bimodal
- High entropy
- High variance

After:

$$p(x) = \delta(x - x^*)$$

17.6. Quantifying the Reduction of Δ

Entropy:

$$\Delta_H^{\text{before}} \approx \log 2 + \varepsilon$$

Variance:

Before:

$$\text{Var}[x] \approx x_0^2$$

After:

$$\text{Var}[x] = 0$$

👉 Reduction:

$$\Delta \rightarrow 0$$

17.7. TNA Interpretation

- The physical system maintains $\Delta > 1$
- The dynamics (N_0) do not select
- N_1 introduces effective uniqueness

👉 Exactly your thesis:

- Physics describes the distribution
 - Observed reality requires selection
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17.8. Strong Conceptual Reading

The double-well reveals something fundamental:

- Multiplicity is not noise
- It is structural

The “randomness” of which well is selected:

- Does not originate from the potential
 - It arises from unresolved selection
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Closing Statement

In a double-well system, physics provides two equally valid futures.
What we observe is not the evolution of the system—
but the consequence of selecting one of them.